DESIGN & BUILD SPEC — FOR CLAUDE CODE

Stream Deck × Claude Monitor

A 15-key Elgato Stream Deck (MK.2) plugin that surfaces live Claude session state and exposes one-press permission actions. This document is the visual + behavioral contract to implement against — colors, icon rules, dynamic states, motion, and per-key detail.

DEVICE	GRID	KEY CANVAS	REPO
Stream Deck MK.2	5 × 3 · 15 keys	72×72 · @2x 144×144	C:\Code\yolodeck

1 · Design principles

Color IS the signal

A glance from across the room must tell you the state. Bright, saturated RGB fills carry meaning; text is confirmation, not the primary read.

Motion means "needs you"

Static keys are steady state. A gentle breathe = Claude is working; a faster pulse = Claude is waiting on you. Nothing else animates.

Solid glyphs, no outlines

Every icon is a single-weight *filled* shape. No stroked/outline icons, no thin hairlines — they disintegrate at 72px behind the key's plastic diffuser.

Dim when irrelevant

Action keys with nothing to act on drop to a dark, desaturated resting state so lit keys stand out. No key is ever fully black unless intentionally empty.

2 · Key canvas & sizing rules

- Render every key at **144×144 px (@2x)** and let Stream Deck downscale; also provide **72×72** for @1x. Ship as PNG (RGBA) via `setImage()`, or draw to an offscreen canvas and pass a data-URL.
- **Full-bleed background.** Paint the whole 144×144 with the state fill — no inner card, no border. The hardware rounds the corners with a ~16px (physical) radius; do not draw your own rounded rect or you'll get a double border. Fill the corners.
- **Safe area:** keep all glyphs and text inside a centered **112×112** region (16px inset @2x). The bezel + diffuser eats the outer ring.
- **Glyph size:** icon-only keys ≈ **72–84px** tall, optically centered. Icon-over-label keys: glyph ≈ 56px, baseline at ~78px, label below.
- **Text:** Inter / SF-style, **600 weight**. Primary label ≥ **22px @2x** (11px @1x) — never smaller. One or two words max. Secondary value line 18px @2x. Never wrap to 3 lines.

- **Contrast:** text is white `#FFFFFF` at 96% on saturated fills; on dim/dark fills use white at 82%. Add a 1px `rgba(0,0,0,.35)` text shadow so labels survive on bright fills.

3 · Color system (state tokens)

Bright, saturated RGB — these are the lit-key fills. Each token is a top→bottom gradient (brighter top edge sells the "lit plastic" look): fill top = listed hex, fill bottom = the hex at 82% lightness. Use flat hex if a gradient is inconvenient.

Idle / resting
`#3E4756 · slate`
Nothing happening; steady.

Working / thinking
`#D97757 · Claude clay`
Breathes slowly (§5).

Awaiting you
`#F5A623 · amber`
Faster pulse (§5).

Approve / go
`#22C55E · emerald`
Approve, Always (active).

Deny / stop / danger
`#EF4444 · red`
Deny (active); yolodeck armed.

Info / neutral-active
`#4AA3F0 · sky`
Model, Focus, nav.

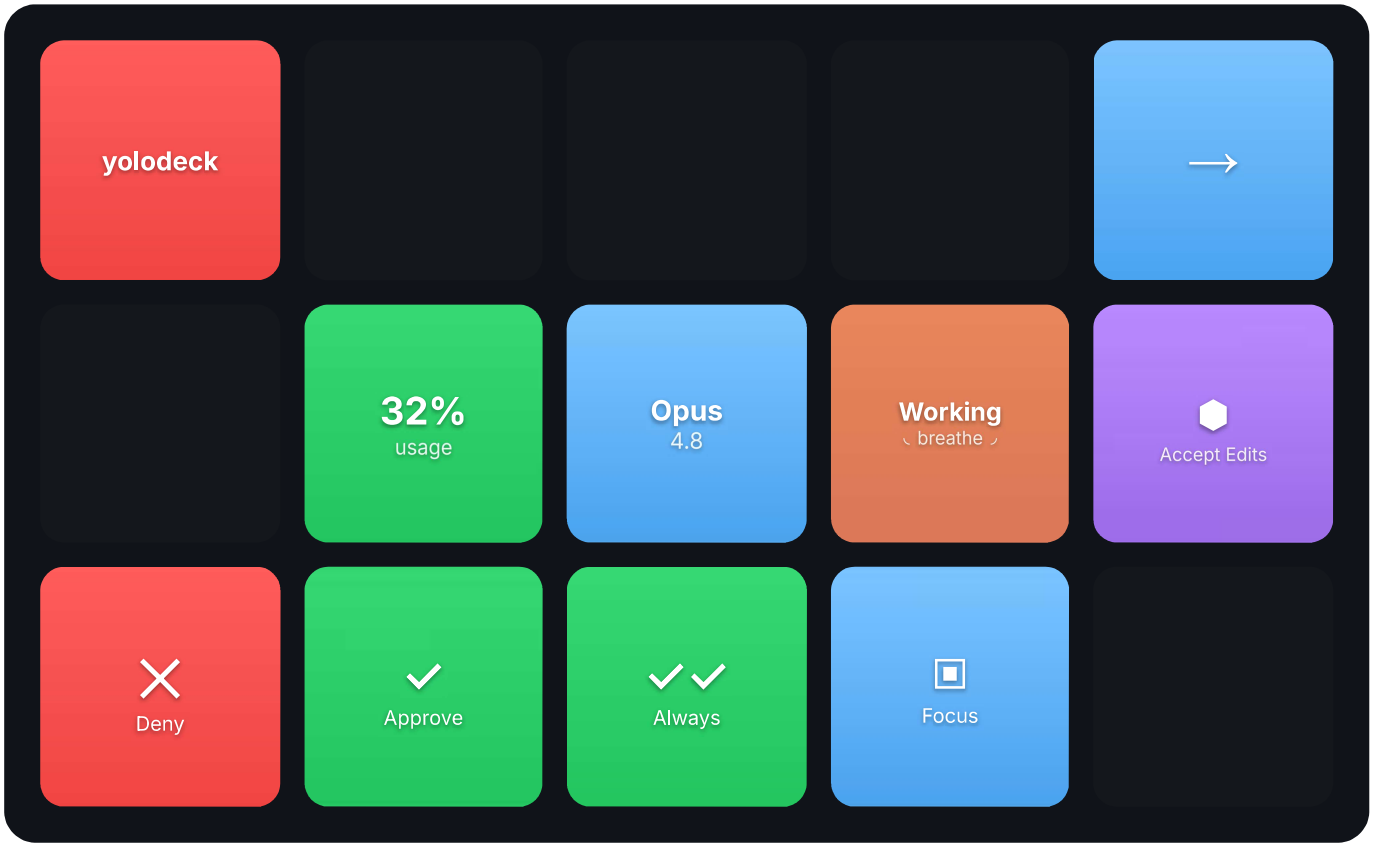
Permission mode
`#9B6BE8 · violet`
Accept-Edits / Plan mode.

Empty / off
`#14171C · near-black`
Unassigned keys only.

Disabled/dim variant of any action color = same hue at **~28% saturation and 40% brightness** (e.g. Approve-dim ≈ `#25412F`). Use for action keys that have no pending request.

4 · Grid layout (5 × 3)

Matches the physical layout in the reference photo. Coordinates are `(row, col)`, 0-indexed from top-left.



Approximation for layout only — final glyphs follow §3 (solid, no outline) and the per-key table in §6. Action keys (Deny/Approve/Always) shown here in their *active* state.

5 · Motion (breathe / pulse)

Stream Deck has no CSS — animate by pushing a new frame on a timer. Drive a single global ticker (~**20 fps**) and re-render only the keys whose state animates. Modulate **overall brightness/opacity of the fill** (multiply RGB by the eased value); keep glyph + text at constant full brightness so they stay legible.

MOTION	USED FOR	PERIOD	RANGE	EASING
Breathe (slow)	Working / thinking	2400 ms	brightness 0.62 ↔ 1.0	sine (ease-in-out)
Pulse (fast)	Awaiting you — Approve/Deny/Always active	1100 ms	brightness 0.55 ↔ 1.0	sine
Flash (once)	Key pressed / action confirmed	180 ms	white flash → settle	ease-out
Steady	Everything else	—	constant	—

Only **one attention motion** should run at a time in the common case (Working breathes, OR the action trio pulses). If both a permission request and background work are true, the pulsing action trio wins the user's attention; keep Working steady-lit meanwhile. Stop all timers when no key needs motion (CPU + LED wear).

6 · Per-key specification

yolodeck

`(0,0) · title / arm toggle`

Branding + master "yolo mode" toggle (auto-approve everything). **OFF:** Idle slate, small "yolodeck" wordmark. **ARMED:** Deny-red fill, wordmark in white, subtle steady glow (no motion) so it reads as a live warning. Press toggles; show 180ms flash on press.

usage

`(1,1) · live data · read-only`

Big number = % of the current usage window consumed; small "usage" label under it. **Color ramps by value:** Approve-green <50%, amber 50–80%, red >80%. **No data:** Idle slate, big "—", label "no data" (as in photo). Optionally press to cycle secondary readout (%, tokens, reset-in). Steady, no motion.

model

`(1,2) · live data`

Current model — family on line 1 ("Opus"), version on line 2 ("4.8"). Info-sky fill, steady. If press cycles model, flash on press then re-render with the new name. If unknown: Idle slate + "—".

status

`(1,3) · live data · the heartbeat`

Idle: slate, "Idle", steady. **Working:** Claude-clay, "Working", *slow breathe*. **Awaiting:** amber, "Waiting", *fast pulse* (mirrors the action trio). **Done:** green, "Done", steady for ~3s then decays to Idle. **Error:** red, "Error", steady. This is the single most-glanced key — make its color transitions instant.

permission mode

`(1,4) · live + toggle`

Solid shield glyph + mode label. Fill encodes the mode: **Default** = slate, **Accept Edits** = violet, **Plan** = sky, **Bypass/yolo** = red. Press cycles modes; flash on press. Keep the label to one word where possible ("Edits", "Plan").

Deny · Approve · Always

`(2,0)(2,1)(2,2) · the action trio`

The one dynamic behavior you called out. **No pending request** → all three *dim* (desaturated dark variant of their hue), no motion, non-actionable feel. **Request pending** → all three light to full saturation and *pulse together* (§5): Deny red ×, Approve green ✓, Always green ✓✓ (double-check, solid). On press: 180ms white flash, then immediately drop the whole trio back to dim (request resolved). Icons are solid glyphs, ~72px, no outline.

Focus

`(2,3) · action`

Brings the Claude/terminal window to the foreground. Solid window glyph + "Focus" label, Info-sky, steady. Flash on press. Optional: tint amber while a request is pending to hint "come look".

nav arrow `(0,4) · navigation`

Solid right-arrow glyph, Info-sky, steady. Reserved for a future second page even though scope is single-page now — keep it as a styled, pressable placeholder rather than blank so the layout reads intentionally.

empty keys `(0,1)(0,2)(0,3)(1,0)(2,4)`

Off / near-black `#14171C`, no glyph, not pressable. Do not leave the hardware default gray.

7 · Rendering & implementation notes

- **One render function.** `renderKey(spec, brightness=1)` draws a 144×144 canvas: fill gradient × brightness → glyph → label → optional text-shadow, returns a PNG data-URL for `setImage()`. Every state is just a different spec object; motion is the same spec at varying brightness.
- **Central store → keys.** Keep one session-state object (status, model, usage, permissionMode, pendingRequest, yoloArmed). On any change, diff and re-render only affected keys. The action trio, status key, and Focus tint all derive from `pendingRequest`.
- **Animation loop.** Single `setInterval(~50ms)`; compute eased brightness from `performance.now()` per motion token; push frames only to animating keys. Clear the interval when nothing animates.
- **Press feedback** is mandatory on every actionable key (the 180ms white flash) — the plastic gives no travel feedback, so the pixels must.
- **Cache** static-state PNGs (idle, dim action keys, empty, model) so you're not re-rasterizing constant frames.
- **Fonts:** bundle the label font with the plugin (don't rely on OS fonts). Inter or SF Pro Display, 600, is the reference.

End of spec · hand to Claude Code alongside the reference photo · questions → iterate on §3 tokens and §6 key cards first.